SYSTEM REQUIREMENTS DOCUMENT

Smart Pharma System for Production & Workforce Management

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Status	Production-Ready
Prepared By	Smart Pharma System for Production & Workforce Management
Document Type	System Requirements (PDM Methodology)

Approval Table

Approver Name	Title	Signature	Date
Project Guide	Supervisor		
Team Leader	Developer		
Client Representative	Pharma Manager		

Section 1 – Purpose

The purpose of this System Requirements document is to define the overall requirements for the Smart Pharma System. The system is developed to improve pharmaceutical production management, workforce tracking, attendance monitoring, barcode verification, packaging management, and reporting processes.

The project provides a centralized digital platform for pharmaceutical industries to automate daily production operations, reduce manual errors, improve monitoring, and increase operational efficiency.

Section 2 – General System Requirements

2.1 Major System Capabilities

The Smart Pharma System must provide the following capabilities:

- System must be available through web browsers.
- System must support 24/7 operation.
- System must support barcode scanning functionality.
- System must provide secure login authentication.
- System must support role-based access control.
- System must provide real-time production monitoring.
- System must generate reports automatically.
- System must support attendance tracking.
- System must support mobile and desktop accessibility.
- System must maintain centralized database storage.
- System must support admin dashboard analytics.
- System must allow task allocation and tracking.

2.2 Major System Conditions

The Smart Pharma System must satisfy the following conditions:

- System must use MongoDB database management system.
- System must operate in a secure network environment.
- System must support barcode camera integration.
- System must support multiple user access simultaneously.
- System must maintain backup and recovery support.
- System must support internet and local network deployment.
- System must maintain data consistency and integrity.
- System must support future AI integration modules.
- System must work with modern web browsers.

2.3 System Interfaces

The Smart Pharma System interfaces with several internal and external components.

Internal Interfaces:

- Employee Management Module
- Attendance Module
- Production Management Module
- Packaging Management Module
- Reporting System

External Interfaces:

- Barcode Scanner Device
- Camera Scanner API
- MongoDB Database Server
- Web Browser Interface

Interface Flow:

User → Frontend Interface → Backend Server → Database

Barcode Scanner → Backend Validation → Production Database

2.4 System User Characteristics

User Type	Function	Device Type	Usage
Admin and ceo	System managemen/ Attendance & salary	Desktop/Laptop	Full access
Manager	Production monitoring	Desktop/Tablet	Monitor tasks
Worker	Task execution	Mobile/Desktop	Scan barcode and update tasks

Estimated Users:

- Admin and ceo: 2–5 users
- Supervisors: 10–20 users
- Workers: 100+ users

Section 3 – Policy and Regulation Requirements

3.1 Policy Requirements

- System must maintain employee confidentiality.
- System must follow organizational data policies.

- System must maintain role-based user permissions.
- System must maintain secure access control.
- System must provide audit tracking for system activities.
- System must maintain backup policies for critical data.

3.2 Regulation Requirements

- System must follow pharmaceutical manufacturing standards.
- System must comply with data security regulations.
- System must support secure employee information handling.
- System must maintain production tracking records.
- System must support barcode compliance standards.

Section 4 – Security Requirements

The Smart Pharma System must include the following security requirements:

- Secure username and password authentication.
- Encrypted database storage.
- Role-based access control.
- Session timeout management.
- Admin-only access for sensitive modules.
- Backup and recovery support.
- Protection against unauthorized access.
- Secure API communication.
- Activity logging and monitoring.
- Data validation and input filtering.

Section 5 – Training Requirements

The following training requirements are necessary:

- Admin training for dashboard management.
- Supervisor training for task monitoring.
- Worker training for barcode scanning.
- HR training for attendance management.
- System maintenance training for technical staff.
- User manual and documentation support.
- Demo sessions for system usage.

Section 6 – Initial Capacity Requirements

The Smart Pharma System must support the following capacity requirements:

- 100–500 active users initially.
- 1000+ daily production transactions.
- Real-time attendance updates.
- Daily barcode scanning operations.
- Monthly report generation.
- Storage support for employee and production records.
- High-speed database access.
- Expandable server storage for future growth.

Estimated Usage:

Component	Estimated Capacity
Employees	500+
Daily Attendance Records	1000+
Barcode Transactions	2000+
Production Tasks	500+ daily
Reports Generated	100+ monthly

Section 7 – Initial System Architecture

The Smart Pharma System architecture includes frontend, backend, database, and hardware integration.

Hardware Requirements:

- Desktop/Laptop systems
- Barcode scanner devices
- Web server system
- Network connectivity devices

Software Requirements:

- Windows/Linux Operating System
- MongoDB Database Server
- XAMPP/WAMP Server
- Web Browser

Programming Languages & Tools:

Component	Technology
Frontend	HTML, CSS, JavaScript, React.js
Backend	NODE.JS / express.js
Database	MongoDB
API	REST API
Barcode Scanner	

Operating Environment:

- Windows 10/11
- Linux Server
- Cloud/Local Server Deployment

System Architecture Flow:

Frontend → Backend → Database

Barcode Scanner → Backend Validation → Database Storage

Section 8 – System Acceptance Criteria

The Smart Pharma System will be accepted when the following conditions are satisfied:

- User login system functions correctly.
- Barcode scanning works without errors.
- Attendance records are stored accurately.
- Production task allocation functions properly.
- Reports are generated successfully.
- Database operations work correctly.
- Dashboard displays real-time information.
- System supports multiple users simultaneously.
- Backup and recovery processes function properly.
- System passes testing and validation.

Section 9 – Current System Analysis

Currently, many pharmaceutical production activities are managed manually using paper-based records and spreadsheets.

Current Problems:

- Manual attendance tracking
- Difficult production monitoring
- Human errors in packaging
- Slow report generation
- Lack of centralized monitoring
- Inefficient employee management

Proposed System Improvements:

- Automated attendance tracking
- Barcode-based product verification
- Real-time monitoring dashboard
- Centralized database management
- Faster report generation
- Improved production accuracy

Section 10 – References

Document No.	Document Title	Date	Author
REF-01	Smart Pharma Project Proposal	2026	Project Team
REF-02	MongoDB Documentation	2026	Oracle
REF-03	React.js Documentation	2026	Meta
REF-04	NODE.JS Documentation	2026	NODE.JS Group

Section 11 – Glossary

Term	Definition
API	Application Programming Interface
DBMS	Database Management System
Barcode	Machine-readable product code
Dashboard	Visual management interface
Authentication	User identity verification
Attendance Module	Employee tracking system
Production Module	Task management system

Section 12 – Document Revision History

Version	Date	Name	Description
1.0	08-05-2026	Project Team	Initial creation of System Requirements document

Section 13 – Appendices

Appendix A – Barcode Example

Example Product Barcode:

BLISTER-10CM-14KG-1001-V1

Appendix B – Future Enhancements

Future improvements include:

- AI-based production prediction
- Machine learning analytics
- IoT sensor integration
- Face recognition attendance
- Cloud deployment
- Mobile application support
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